

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Module - II

Practical – XII

Roll No.:T054	Name:Priti Yadav
Class:TYIT	Batch:2
Date of Assignment: 3/10/26	Date/Time of Submission: 3/10/26

12. Analysis of Variance and Experimental Design:

a. Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance(ANOVA), and draw conclusions.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway

# 1. Load dataset
df = pd.read_csv("Retail and warehouse Sale.csv")

# 2. Display first five records and shape
print(df.head())
print("Rows and Columns:", df.shape)

# 3. Display column names and data types
print("\nColumn Names:")
print(df.columns)
print("\nData Types:")
print(df.dtypes)

# 4. Check missing values and duplicates
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Records:")
print(df.duplicated().sum())

# 5. Display ITEM TYPE categories
print("\nITEM TYPE:")
print(df["ITEM TYPE"].unique())

# 6. Average RETAIL SALES for each ITEM TYPE
average_sales = df.groupby("ITEM TYPE")["RETAIL SALES"].mean()
print("\nAverage Retail Sales:")
print(average_sales)

# 7. Box plot
plt.figure(figsize=(10,5))
sns.boxplot(x="ITEM TYPE", y="RETAIL SALES", data=df)
```

```
plt.title("Retail Sales by Item Type : Priti Yadav T054")
plt.xticks(rotation=45)
plt.show()

# 8. Hypotheses
print("\nH0: Average retail sales are equal among all item types.")
print("H1: Average retail sales are different among item types.")

# 9. One-way ANOVA
groups = []
for item in df["ITEM TYPE"].unique():
    groups.append(df[df["ITEM TYPE"] == item]["RETAIL SALES"])

f, p = f_oneway(*groups)

# 10. F-statistic and p-value
print("\nF-statistic:", f)
print("p-value:", p)

# 11 & 12. Decision
if p < 0.05:
    print("Reject H0")
else:
    print("Do not reject H0")

# 13. Conclusion
if p < 0.05:
    print("There is a significant difference in retail sales among item types.")
else:
    print("There is no significant difference in retail sales among item types.")
```

OUTPUT:

```
YEAR  MONTH  SUPPLIER ITEM CODE \
0  2020      1  REPUBLIC NATIONAL DISTRIBUTING CO  100009
1  2020      1                PWSN INC  100024
2  2020      1      RELIABLE CHURCHILL LLLP  1001
3  2020      1  LANTERNA DISTRIBUTORS INC  100145
4  2020      1      DIONYSOS IMPORTS INC  100293

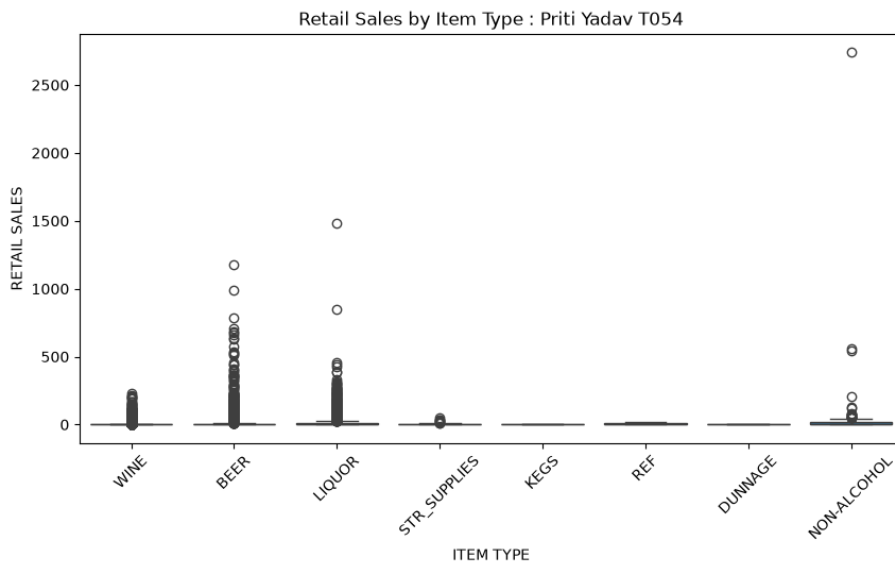
ITEM DESCRIPTION ITEM TYPE  RETAIL SALES \
0  BOOTLEG RED - 750ML  WINE  0.00
1  MOMENT DE PLAISIR - 750ML  WINE  0.00
2  S SMITH ORGANIC PEAR CIDER - 18.7OZ  BEER  0.00
3  SCHLINK HAUS KABINETT - 750ML  WINE  0.00
4  SANTORINI GAVALA WHITE - 750ML  WINE  0.82

RETAIL TRANSFERS  WAREHOUSE SALES
0  0.0  2.0
1  1.0  4.0
2  0.0  1.0
3  0.0  1.0
4  0.0  0.0
Rows and Columns: (30000, 9)
```

```

Column Names:
Index(['YEAR', 'MONTH', 'SUPPLIER', 'ITEM CODE', 'ITEM DESCRIPTION',
      'ITEM TYPE', 'RETAIL SALES', 'RETAIL TRANSFERS', 'WAREHOUSE SALES'],
      ...
REF          5.783750
STR_SUPPLIES 5.485714
WINE         3.195334
Name: RETAIL SALES, dtype: float64

```



H0: Average retail sales are equal among all item types.
H1: Average retail sales are different among item types.

F-statistic: nan
p-value: nan
Do not reject H0
There is no significant difference in retail sales among item types.

b. Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.

CODE:

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.formula.api import ols

# Load dataset
df = pd.read_csv("Retail and warehouse Sale.csv")
# Remove missing values
df = df.dropna(subset=["ITEM TYPE", "YEAR", "RETAIL SALES"])
# 14. Select ITEM TYPE, YEAR and RETAIL SALES
data = df[["ITEM TYPE", "YEAR", "RETAIL SALES"]]
print("Selected Data:")
print(data.head())
# Check number of years
print("\nYears available:")
print(data["YEAR"].unique())
# 15. Average retail sales for each ITEM TYPE and YEAR
average_sales = data.groupby(
    ["ITEM TYPE", "YEAR"]
)[["RETAIL SALES"].mean()
print("\nAverage Retail Sales:")
print(average_sales)
# 16. Visualization
plt.figure(figsize=(10, 5))
sns.barplot(
    data=data,
    x="ITEM TYPE",
    y="RETAIL SALES")
plt.title("Average Retail Sales by Item Type : Priti Yadav T054")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
# Check whether YEAR has more than one value
if data["YEAR"].nunique() < 2:
    print("\nFactorial ANOVA cannot be performed.")
    print("Reason: The dataset contains only one YEAR (2020).")
    print("Therefore, the effect of YEAR and ITEM TYPE × YEAR")
    print("interaction cannot be calculated.")
else:
    # 17, 18, 19 and 20. Factorial ANOVA
    model = ols(
        "Q('RETAIL SALES') ~ C(Q('ITEM TYPE')) * C(YEAR)",
        data=data
    ).fit()
    anova = sm.stats.anova_lm(model, typ=2)
    # 21. Display results
    print("\nFactorial ANOVA Results:")

```

```

print(anova)
# 22. Check significance
print("\nSignificance:")
for effect in anova.index:
    if effect != "Residual":
        p = anova.loc[effect, "PR(>F)"]
        if p < 0.05:
            print(effect, "is Significant")
        else:
            print(effect, "is Not Significant")
# 23. Conclusion
print("\nConclusion:")
print("Factorial ANOVA was performed successfully.")

```

OUTPUT:

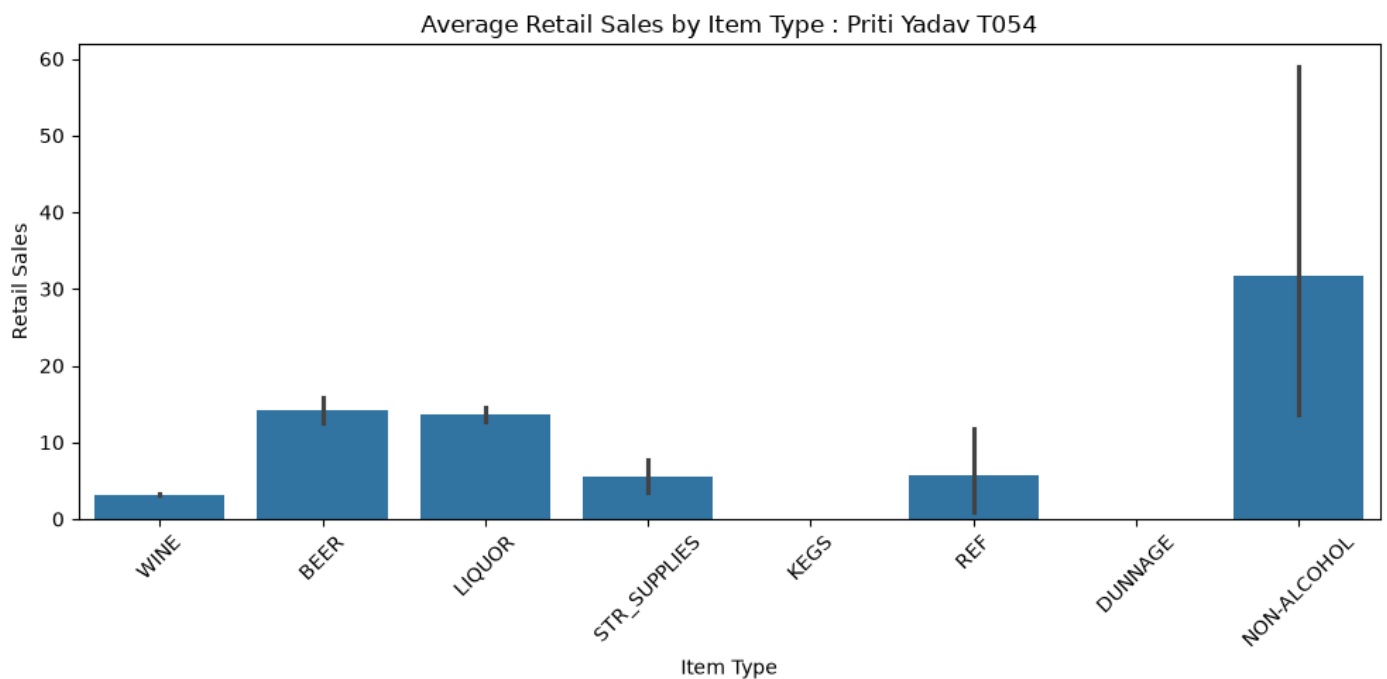
```

Selected Data:
  ITEM TYPE  YEAR  RETAIL SALES
0    WINE  2020      0.00
1    WINE  2020      0.00
2    BEER  2020      0.00
3    WINE  2020      0.00
4    WINE  2020      0.82

Average Retail Sales:
  ITEM TYPE  YEAR  RETAIL SALES
BEER      2020  14.118748
DUNNAGE   2020   0.000000
KEGS      2020   0.000000
LIQUOR    2020  13.635171
NON-ALCOHOL 2020  31.742419
REF       2020   5.783750
STR_SUPPLIES 2020   5.485714
WINE      2020   3.195334
Name: RETAIL SALES, dtype: float64

Years available:
[2020]

```



Factorial ANOVA cannot be performed.
Reason: The dataset contains only one YEAR (2020).
Therefore, the effect of YEAR and ITEM TYPE × YEAR
interaction cannot be calculated.